Ghost Chatbot: A Multi-Game-Type Narrative Approach to Teaching IBM SkillsBuild Chatbot/NLP Concepts

Preliminary Project Report

7CCSMPRJ Individual Project

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1 Introduction

1.1 Background and Motivation

The rapid adoption of conversational AI across enterprise, healthcare, and public services has generated significant demand for practitioners who understand the principles of chatbot design. IBM SkillsBuild's course *Build Your First Chatbot Using IBM watsonx* provides a structured vocational curriculum covering Intent classification, Entity extraction, Dialog Node configuration, Confidence scoring, Fallback design, and system integration. However, substantial parts of the delivery format remain passive: video lectures, static text, and predefined multiple-choice assessments. Research on multimedia learning suggests that such formats produce shallow rather than transferable understanding in technical domains, particularly for learners without prior programming experience [9].

Game-based learning (GBL) has been proposed as a theoretically grounded alternative for complex, skill-based domains [13]. Educational games are designed to situate abstract concepts within meaningful problem contexts, require learners to apply knowledge through consequential decisions, and provide immediate low-stakes feedback. Despite this evidence base, the intersection of GBL and vocational chatbot design literacy has received limited attention in the published literature.

This project responds to that gap by designing *Ghost Chatbot*, a narrative puzzle game in which the IBM SkillsBuild chatbot curriculum is covered across 24 specified levels using a variety of puzzle mechanic types, each intrinsically tied to the challenge structure of its target NLP concept. A playable prototype covering one representative level per Act will be implemented for evaluation purposes. The player trains an AI entity, Ghost, which is trapped in a smart home system, and guides its communicative ability from system menus and emoji to natural human speech in direct correspondence with the NLP concepts operationalised in each Act. A postdoctoral researcher character, Lily, powered by a locally hosted large language model (IBM Granite via Ollama), will provide tiered scaffolded dialogue in prototype contexts.

1.2 Problem Statement

Three interrelated problems motivate this work.

Problem 1: Substantial parts of the IBM SkillsBuild chatbot curriculum are delivered through passive instruction. Concepts such as Intent, Entity, Dialog Node, and Confidence Score are often presented through video demonstrations and predefined questions rather than through design decisions with observable consequences. The design-reasoning competencies required for vocational chatbot literacy are not developed through this approach [9, 15].

Problem 2: Published interventions using a multi-game-type approach to operationalise vocational chatbot concepts appear to be scarce. While GBL has been applied to programming, mathematics, and science, to the best of the author's knowledge no game-based intervention targeting chatbot design concepts as the primary subject of learning has been identified in the literature. In existing interventions, structurally distinct puzzle mechanics are not employed for different concept types. Block-based environments and platform-guided labs are oriented toward procedural familiarity rather than system-design reasoning [12].

Problem 3: In existing pedagogical agent research, an agent's communicative capability is not coupled mechanically to the player's completion of concept-specific in-game tasks. Existing work examines agent appearance, voice, and expertise style [10, 7]. In this project, Ghost's expressive range, from emoji to natural speech, is coupled to the player's completion of concept-specific in-game tasks, so that Ghost's changing speech ability makes the game's concept progression visible to the player.

1.3 Research Approach

A design-based research (DBR) paradigm is adopted [1]. The primary research output is the designed artefact, *Ghost Chatbot*, in which three theoretical mechanisms are embodied: game-based learning, narrative-centred learning environments (NCLE), and pedagogical agent (PA) theory. Under DBR, the research question is directed at the artefact's internal validity and system-level correctness rather than at learner outcomes.

The design validity and system behaviour of *Ghost Chatbot* are evaluated in this study; learning outcomes are not. No user study is conducted. This reflects the practical scope of an MSc-level project, in which a full empirical evaluation of learning outcomes would require ethical approval, participant recruitment, and controlled conditions that are not feasible within the project timeline. Whether *Ghost Chatbot* produces measurable learning gains is identified as a priority direction for future empirical work.

1.4 Research Question

To what extent does a multi-game-type narrative game design operationalise selected IBM SkillsBuild chatbot/NLP concepts through intrinsically integrated puzzle mechanics within a playable prototype?

This question is answered through three design and system analyses conducted without user participants (Table 1).

Evaluation focus	Method	Evidence produced
Curriculum operationalisation and intrinsic integration	Curriculum mapping matrix; NCLE-inspired design checklist [14]; subset consistency check (where feasible)	IBM concept-to-level matrix; intrinsic/partial/extraneous integration judgements with written justification
LLM scaffolding behaviour	Fixed prompt bank (one per Act, three hint tiers; each prompt submitted three times); three-dimension rubric	Rubric scores per evaluated Act and tier; qualitative failure mode categories
Narrative-pedagogical coupling	Document analysis of Game Design Document (GDD)	Concept-to-capability correspondence table for Acts 0–6; preliminary consistency analysis

Table 1: Evaluation foci, methods, and evidence.

1.5 Technical Specifications

- **Game client:** Unity 6 (C#), WebGL build target for local browser-based demonstration; single-player
- **Puzzle mechanics:** Eight structurally distinct interaction types: drag-and-drop classification, span annotation, node assembly, slider calibration, flow diagram construction, multiple choice, form configuration, and ordered list
- **Prototype scope:** A 24-level curriculum is fully specified in the GDD; a playable prototype implementing one complete level per Act (8 levels) will be developed for evaluation
- **AI tutor:** IBM Granite via Ollama, accessed through a local Node.js proxy during prototype evaluation; constrained by a curriculum-specific system prompt
- **Backend:** Node.js with TypeScript; REST API used for progress synchronisation and attempt logging
- **Database:** SQLite for prototype development and local evaluation
- **Progress profile:** A local pseudonymous profile will be used to associate prototype progress and attempt logs with a test session
- **Analytics:** Puzzle attempt logs and hint triggers will be recorded for internal prototype validation

1.6 Contributions

Design contribution: The intended design contribution is a fully specified 24-level curriculum-to-game design and an 8-level playable prototype, in which each puzzle mechanic will be intrinsically integrated with a specific chatbot design concept.

Conceptual contribution: A character-as-system coupling mechanism is proposed, in which a pedagogical agent's communicative capability evolves in direct mechanical correspondence with the player's completion of concept-specific in-game tasks, extending prior pedagogical agent research [10, 7].

Methodological contribution: The intended methodological contribution is a structured approach for evaluating design-based learning systems through curriculum mapping, system testing, and design analysis prior to empirical evaluation.

Scope boundary: This project is scoped to the IBM SkillsBuild chatbot/NLP curriculum. Whether the multi-game-type framework generalises to other IBM vocational curricula is identified as a direction for future work, as different curricula may require structurally different mechanic types.

2 Background and Literature Review

2.1 Game-Based Learning and Intrinsic Integration

Game-based learning has been proposed as an approach in which abstract knowledge is situated within goal-directed tasks with observable consequences [13]. Mayer (2019) reviews evidence on computer games in education and argues that claims about educational games should be tested through rigorous research and grounded in evidence-based theories of learning. For this project, the key design implication is that game mechanics should embody the target knowledge rather than merely decorate instructional content. In *Ghost Chatbot*, each puzzle mechanic is designed such that solving the puzzle constitutes the act of demonstrating understanding of the target concept. Sweller's (1988) cognitive load theory is also relevant here: intrinsic cognitive load may exceed working memory capacity in complex technical domains, and this risk is addressed through progressive complexity, with one new conceptual component introduced per Act.

2.2 Narrative-Centred Learning Environments

A narrative-centred learning framework is proposed by Lester et al. (2014), in which story provides the semantic context within which new concepts acquire meaning. Rowe et al. (2011) examine narrative-centred learning environments as systems that integrate learning, problem solving, and engagement. In this project, this perspective is operationalised through evaluation criteria covering narrative-driven goals, responsive characters, problem-solving episodes embedded in the story world, and assessment that is presented as natural game feedback. Narrative integration is applied throughout *Ghost Chatbot* such that educational concepts function as plot mechanisms: Ghost is unable to communicate beyond system menus in Act 0 because Intent classification has not been trained, and contextual memory is not retained in early Act 3 because Context Variables have not been implemented.

2.3 Pedagogical Agents and Character-as-System Coupling

Moreno et al. (2001) reported evidence that pedagogical agent-mediated instruction can produce deeper learning than equivalent text-only instruction through social agency effects. Kim and Baylor (2006) distinguish companion-style agents from expert tutors, noting that companion agents may support motivation and self-efficacy under certain conditions. In *Ghost Chatbot*, Lily is designed as a companion character who does not disclose puzzle solutions. Ghost represents an extension of existing agent research: rather than exhibiting a fixed personality, Ghost's communicative capability evolves in direct mechanical correspondence with the NLP concepts operationalised by the player's in-game task progression. This character-as-system coupling design has received limited explicit treatment in the published literature.

2.4 Chatbot Education and the Design Literacy Gap

Winkler and Söllner (2018) survey chatbot applications in education and identify common usage patterns such as FAQ systems and content delivery agents, but their focus is on chatbots as educational tools rather than chatbot design and training as the learning subject. Pérez et al. (2020), reviewing 80 studies, confirm the dominance of chatbots as delivery tools and identify the teaching of chatbot design concepts as an area with limited prior investigation. Graesser et al. (2001) identify conversational dialogue as a core mechanism in intelligent tutoring systems, where tutors present problems and questions and support learners through multi-turn dialogue, which informs the design of Lily's LLM interaction model. The reviewed approaches share one limitation relevant to this project: chatbot design decisions are rarely turned into playable actions with visible, narratively consequential outcomes.

2.5 Large Language Models as Educational Scaffolding

Kasneci et al. (2023) outline opportunities and risks of large language models in education, including personalised feedback and hallucination. Huber et al. (2024) argue that GBL can mitigate LLM over-reliance risk, as games require active decision-making that conversational LLMs cannot enforce; this argument supports the design choice to constrain Lily from providing direct puzzle solutions. Goslen et al. (2025) provide empirical support for LLM-based student plan generation intended to support adaptive scaffolding in the Crystal Island game-based learning environment, within a related narrative-centred game-based learning tradition. Gong et al. (2026) demonstrate learning gains and reduced cognitive load under LLM scaffolding in a game-based AI literacy setting. As Nye et al. (2023) note, prompt constraints play an important role in shaping tutor-appropriate LLM behaviour; accordingly, Lily's system prompt is curriculum-specific and prohibits answer disclosure.

2.6 Theoretical Integration

These three mechanisms, GBL, NCLE, and pedagogical agent design, are treated as interdependent in this project. Narrative provides the motivation that sustains engagement with the puzzles; the puzzles provide the design decisions that give Ghost's growth its semantic meaning; and Ghost's growth provides the feedback that makes narrative investment educationally productive. The design therefore depends on the interaction between mechanics, narrative, and agents, rather than on any one element in isolation. The three evaluations are designed to assess whether each mechanism has been correctly operationalised within the integrated design.

3 Proposed System and Evaluation Methodology

3.1 System Overview

Ghost Chatbot is designed as a single-player narrative puzzle game in which the IBM SkillsBuild chatbot curriculum is operationalised through structurally distinct puzzle mechanics across 24 levels. A design-based research methodology is adopted [1]: requirements for the system are derived from the theoretical literature and the IBM SkillsBuild course content, and evaluation is directed at whether those requirements are satisfied in the artefact. The final report will present a requirements traceability matrix linking literature-derived design principles, IBM course concepts, implemented mechanics, and evaluation evidence.

A playable prototype covering one complete level per Act (8 levels) will be implemented for evaluation. The full 24-level design is specified in the Game Design Document (GDD). The Acts and their corresponding NLP topics are as follows:

- **Act 0:** Chatbot fundamentals (definitions, rule-based vs. AI-enabled, five components, four challenges)
- **Act 1:** Intent (classification, training examples)
- **Act 2:** Entity (extraction, synonyms, system vs. custom entities)
- **Act * (supplementary):** NLP pipeline, tokenisation, POS tagging, named entity recognition, sentiment analysis; implemented in the prototype but excluded from the primary Evaluation 2 and Evaluation 3 scope
- **Act 3:** Dialog (nodes, branching, slots, response types, context variables)
- **Act 4:** Confidence and fallback (scoring, threshold calibration, disambiguation)
- **Act 5:** Testing and debugging
- **Act 6:** Integration and deployment (represented as a deployment design puzzle, not a production chatbot deployment)

3.2 Evaluation Design

The research question is answered through three complementary analyses as summarised in Table 1. No user study is conducted and no participant recruitment is required.

The curriculum mapping matrix (Evaluation 1) is constructed by mapping each IBM SkillsBuild learning objective to the corresponding level in the full 24-level GDD, and assessing whether the puzzle mechanic requires the player to apply the concept (intrinsic) or merely observe it (extraneous), following the criterion established by Mayer (2019). Implementation evidence from the 8-level prototype is then used to confirm whether the representative levels realise the mapped mechanics in practice. The LLM prompt bank (Evaluation 2) consists of one test prompt per Act across three hint tiers, producing 21 test cases in total (Acts 0–6; Act * is excluded from the primary evaluation scope as an optional supplementary Act). Each test case will be submitted three times to account for output variability, resulting in 63 LLM responses for rubric scoring across curriculum relevance, scaffolding tier adherence, and factual accuracy. For Evaluation 3, a preliminary analysis has been conducted against the current GDD. The final report will present the full correspondence table and identify strong, partial, or weak mappings across the 21 Act 0–6 correspondences.

3.3 LLM Scaffolding Architecture

Lily's responses are generated by IBM Granite via Ollama, a locally hosted large language model. The model is constrained by a curriculum-specific system prompt that restricts responses to the current Act's concept scope and prohibits direct disclosure of puzzle solutions. A hint tier system governs response depth: at tiers 1 and 2, Socratic prompting is used; at tier 3 and above, direct explanation is permitted. These constraints form the basis of Evaluation 2.

4 Project Schedule

The project is planned across seven phases between April 2026 and August 2026, as summarised in Table 2.

Work package	Apr	May	Jun	Jul	Aug
Preliminary report	X				
Unity core infrastructure		X			
Prototype level implementation		X	X	X	
Backend and LLM integration			X	X	
Evaluation				X	X
Final report writing			X	X	X
Final build and submission					X

Table 2: High-level project timeline (April–August 2026).

Phase 1: Preliminary report (by 30 April 2026)

Phase 2: Unity core infrastructure (May 2026) — Core UI components will be developed (DropZone, DialogueBox, ChoiceButton, SliderPuzzle, FormPanel, OrderableList), the Ghost system and HintSystem will be implemented, and scene management will be set up. DraggableNode is already complete.

Phase 3: Prototype level implementation (May to July 2026) — One complete level will be implemented per Act (8 levels in total). Priority will be given to Acts 0, 1, and 2, as these introduce the most commonly reused mechanic types. Target: 5 July 2026.

Phase 4: Backend and LLM integration (June to July 2026) — A Node.js and SQLite backend will be set up with REST API endpoints. IBM Granite via Ollama will be integrated and the Lily system prompt will be configured. Unity will be connected to the backend via ApiClient. Target: 19 July 2026.

Phase 5: Evaluation (July to early August 2026) — Evaluation 1 (curriculum mapping) will be conducted against the full GDD. Evaluation 2 (LLM prompt bank) will be conducted after backend integration. A preliminary version of Evaluation 3 has already been completed. Target: 2 August 2026.

Phase 6: Final report writing (June to August 2026) — Chapters 1 to 3 are substantially drafted. Chapters 4 to 6 will be written after evaluation is complete. Target: 9 August 2026.

Phase 7: Final build and submission (by 16 August 2026) — A WebGL build will be produced, the code repository will be prepared, and the final report will be submitted via KEATS.

Prototype scope: The target prototype covers one level per Act (8 levels). The guaranteed minimum covers five structurally distinct mechanics: drag-and-drop classification, slider calibration, node assembly, span annotation, and multiple choice. If LLM integration is delayed, static hints are used as a gameplay fallback, and the LLM scaffolding evaluation is reported as a limitation or deferred component.

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